

CO 4 - ASSESSMENT TOOL -1

Name : Lanka Kavya
Registration Number : 192472298

Subject Code : CSA1003
Subject Name : Software Engineering

Hotel Reservation and Room Management Platform with Dynamic Pricing

Test Case Design Assignment

Hotel Reservation and Room Management Platform with Dynamic Pricing

1. Problem Statement

The **Hotel Reservation and Room Management Platform with Dynamic Pricing** is a software application designed to manage hotel room reservations, room availability, customer information, and hotel operations.

The system allows customers to register, log in, search for available rooms, view room details, check dynamically calculated prices, and reserve rooms. Hotel administrators can manage rooms, reservations, customers, room status, and pricing rules.

The platform uses a **Dynamic Pricing Module** to adjust room prices according to factors such as demand, occupancy, season, room availability, day of the week, and special events.

2. Functional Requirements

Requirement ID	Functional Requirement
FR01	User should be able to register an account.
FR02	User should be able to log in and log out.
FR03	User should be able to search for available rooms.
FR04	User should be able to view room details.
FR05	User should be able to check room prices.
FR06	User should be able to make a room reservation.
FR07	User should be able to cancel a reservation.

3. Non-Functional Requirements

Requirement ID	Non-Functional Requirement
NFR01	System should provide secure user authentication.
NFR02	System should respond quickly to room searches.
NFR03	System should maintain data consistency.
NFR04	System should be available to multiple users simultaneously.
NFR05	System should prevent unauthorized admin access.
NFR06	System should provide a user-friendly interface.
NFR07	System should handle increasing numbers of rooms and reservations.
NFR08	System should maintain reservation reliability.
NFR09	Sensitive information should be protected.
NFR10	Dynamic pricing should provide consistent and accurate results.

4. Test Scenarios

User Management

Room Management

Reservation Management

Dynamic Pricing

Security

5. Detailed Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	Register user with valid data	Open registration → Enter details → Submit	Name: Kavya, valid email, valid password	Account should be created successfully	To be recorded	Pass/Fail
TC02	Register with missing email	Enter name/password → Leave email empty → Submit	Email = Empty	Validation message should be displayed	To be recorded	Pass/Fail
TC03	Register existing user	Enter already registered email	Existing email	System should reject duplicate registration	To be recorded	Pass/Fail

6. Boundary Value Analysis

Boundary Value Analysis is useful for values where behavior can change at the minimum or maximum limits.

Example: Number of Guests

Assume the system allows **1–10 guests**.

Test ID	Input	Expected Result
BVA01	0 guests	Reject
BVA02	1 guest	Accept
BVA03	2 guests	Accept
BVA04	9 guests	Accept
BVA05	10 guests	Accept
BVA06	11 guests	Reject

7. Equivalence Partitioning

The room availability input can be divided into valid and invalid partitions.

Partition	Example	Expected Result
Valid dates	Future check-in and valid checkout	Accepted
Past check-in	Yesterday's date	Rejected
Checkout before check-in	25 Aug → 20 Aug	Rejected
Same dates	25 Aug → 25 Aug	Rejected
Invalid date format	25/ABC/2026	Rejected

This reduces the number of test cases while still covering important input classes.

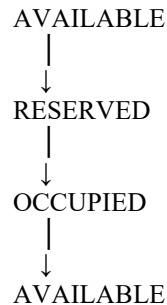
8. Decision Table Testing – Dynamic Pricing

Dynamic pricing can depend on demand and occupancy.

Condition	Low Demand	Normal Demand	High Demand
Low Occupancy	Low Price	Normal Price	Increased Price
Normal Occupancy	Low/Normal Price	Normal Price	Increased Price
High Occupancy	Normal Price	Increased Price	Highest Price

9. State Transition Testing

A hotel room can move through different states.



Another possible transition is:

AVAILABLE → MAINTENANCE → AVAILABLE

State Test Cases

Test ID	Current State	Action	Expected State
ST01	Available	Customer books room	Reserved
ST02	Reserved	Customer checks in	Occupied
ST03	Occupied	Customer checks out	Available
ST04	Available	Admin marks maintenance	Maintenance
ST05	Maintenance	Admin completes maintenance	Available

10. Security Test Cases

Test ID	Security Scenario	Expected Result
SEC01	Login with wrong password	Access denied
SEC02	Customer accesses admin dashboard	Access denied
SEC03	Access protected API without token	Request rejected
SEC04	Empty authentication fields	Validation error
SEC05	Invalid user session	User redirected to login
SEC06	Multiple failed login attempts	System should apply configured security controls
SEC07	Attempt to modify another user's booking	Request should be denied

11. Performance and Reliability Test Cases

Test ID	Scenario	Expected Result
PERF01	Multiple users search rooms simultaneously	System should respond correctly
PERF02	Multiple users attempt same room	Double booking should be prevented
PERF03	Large number of rooms	Search should remain functional
PERF04	Large number of reservations	Reservation history should load correctly
PERF05	Database temporarily unavailable	Appropriate error should be displayed
PERF06	Server restart	Application should recover without corrupting existing data

12. Traceability Matrix

Requirement	Test Cases	Coverage
FR01 User Registration	TC01–TC03	Covered
FR02 Authentication	TC04, TC05, TC25	Covered
FR03 Room Search	TC06, TC07	Covered
FR04 Room Details	TC08	Covered
FR05 Room Pricing	TC15–TC19	Covered
FR06 Reservation	TC09, TC10, TC23	Covered
FR07 Cancellation	TC11	Covered
FR08 Double Booking Prevention	TC23	Covered
FR09 Room Creation	TC12	Covered

13. Test Coverage

The test suite covers:

- User registration and authentication.
- Room search and availability.
- Room management.
- Reservation management.
- Cancellation.
- Double-booking prevention.

Dynamic pricing.

14. Test Design Techniques Used

1. Equivalence Partitioning
2. Boundary Value Analysis
3. Decision Table Testing
4. State Transition Testing

15. Overall Expected Result

After executing the complete test suite, the platform should:

Successfully register and authenticate users.

Display correct room availability.

Prevent booking of unavailable rooms.

Create valid reservations.

Conclusion

The test case design for the **Hotel Reservation and Room Management Platform with Dynamic Pricing** provides comprehensive coverage of functional and non-functional requirements. Normal, boundary, invalid, and exceptional conditions are included along with Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing.

The test cases provide traceability between requirements, scenarios, and expected results, helping verify the correctness, reliability, security, and usability of the platform.

The test design follows the assessment requirements for identifying requirements and scenarios, preparing comprehensive test cases, providing test data and expected results, applying test-design techniques, and establishing test coverage.